ASSISTED BY AI

Proyecto 2

Optimización del padding en LLMs

Grupo SI-35

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1. Implementación de algoritmos de búsqueda local

En esta sección debes describir las particularidades de la implementación de cada uno de los algoritmos de búsqueda local. Es especialmente relevante el proceso de selección de sucesores y actualización del estado actual (o estados actuales en el caso de Beam Search).

1.1. Stochastic Hill Climbing (stochastic-hill-climbing.R)

En este algoritmo la particularidad reside en que el sucesor no se elige el mejor absoluto. En su lugar, el mejor sucesor se elige aleatoriamente de entre una lista de los mejores sucesores que mejoran el estado actual, introduciendo un componente estocástico. Esto permite dotar al algoritmo de cierta capacidad de exploración ante zonas del espacio de búsqueda.

1.2. Random Restart Hill Climbing (random-restart-hill-climbing.R)

Este algoritmo ejecuta el método iterativo básico de Hill Climbing estándar un número n de veces. En cada iteración arranca desde un estado inicial generado aleatoriamente. Al finalizar los n reinicios, el algoritmo compara los resultados de todas las ejecuciones y devuelve como resultado final aquel óptimo local que haya obtenido el menor coste (mejor evaluación).

1.3. Stochastic Random Restart Hill Climbing (stochastic-random-restart.R)

De forma equivalente al anterior, este algoritmo ejecuta iterativamente el algoritmo Stochastic Hill Climbing un número n de veces con distintos estados iniciales. Conserva la memoria del mejor estado encontrado en todas las ejecuciones. La combinación de reinicios aleatorios con decisiones estocásticas aumenta significativamente la componente de exploración del espacio de búsqueda. Ejecuta el algoritmo Stochastic Hill Climbing un número n de veces y se queda con el mejor resultado

1.4. Local Beam Search (local-beam-search.R)

Genera un número k de beams (estados iniciales aleatorios). En cada iteración, se expanden todos los sucesores posibles de los k estados actuales. En lugar de que cada beam siga su propio mejor camino de forma independiente, todos los sucesores se juntan en una lista común. De esa lista global se seleccionan los k mejores estados para conformar los beams de la siguiente iteración. Este enfoque concentra la búsqueda en las regiones más prometedoras garantizando elitismo.

2. Análisis de los resultados

En esta sección debes realizar la comparación de la ejecución de las 11 variantes de algoritmos para las instancias seleccionadas del problema (**basic.txt**, **trade.txt** y **trap.txt**). Las métricas a comparar serán:

Algoritmo	Instancia	Mejor coste	Peor coste	Coste Medio	T. Medio Ejecución
Hill Climbing (básico)	basic.txt	80	80	80.0	~ 5014
	trade.txt	677	797	730.2	~ 8544
	trap.txt	720	4400	2560.0	~ 11734
Stochastic HC	basic.txt	80	80	80.0	~ 9800
	trade.txt	677	1449	819.4	~ 10255
	trap.txt	2560	4400	2928.0	~ 21473
Random Restart (5)	basic.txt	80	80	80.0	~ 4945
	trade.txt	677	705	688.2	~ 10926
	trap.txt	720	2560	1584.0	~ 11027
Random Restart (15)	basic.txt	80	80	80.0	~ 4786
	trade.txt	677	705	682.6	~ 4668
	trap.txt	720	2560	1216.0	~ 10582
Random Restart (30)	basic.txt	80	80	80.0	~ 8385

	trade.txt	677	677	677.0	~ 6381
	trap.txt	720	2560	904.0	~ 8942
Stochastic RR (5)	basic.txt	80	80	80.0	~ 10526
	trade.txt	677	705	682.6	~ 15891
	trap.txt	720	2560	2008.0	~ 32014
Stochastic RR (15)	basic.txt	80	80	80.0	~ 7058
	trade.txt	677	705	682.6	~ 7490
	trap.txt	720	2560	1544.0	~ 16186
Stochastic RR (30)	basic.txt	80	80	80.0	~ 6814
	trade.txt	677	677	677.0	~ 6554
	trap.txt	720	720	720.0	~ 12431
Beam Search (3)	basic.txt	80	80	80.0	~ 17614
	trade.txt	677	765	719.4	~ 13865
	trap.txt	720	2560	2192.0	~ 57161
Beam Search (5)	basic.txt	80	80	80.0	~ 24208
	trade.txt	677	765	711.0	~ 23142
	trap.txt	720	2560	1824.0	~ 92100

Beam Search (10)	basic.txt	80	80	80.0	~ 44781
	trade.txt	677	737	694.0	~ 42365
	trap.txt	720	2560	1824.0	~ 140621

2.1. Resultados Obtenidos

Genera una o varias tablas resumen con las métricas obtenidas para cada una de las instancias evaluadas. (Nota: Se valorará positivamente la inclusión de gráficas, por ejemplo, Tiempo vs. Coste Medio, para ilustrar los resultados).

Hill Climbing

basic.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
10	Local_Best	3046	80	963	1-3-1-5-2-5-3-5-2-3-4-4-5-2-4-1-2-1-3-4	3-3-3-3-5-5-5-5-2-2-2-2-1-1-1-1-4-4-4-4
8	Local_Best	2662	80	7721	3-1-2-2-1-2-4-4-3-5-4-1-4-3-3-5-5-2-1-5	2-2-2-2-4-4-4-4-1-1-1-1-3-3-3-3-5-5-5-5
8	Local_Best	2662	80	6887	3-4-3-3-5-2-4-5-1-2-4-1-3-5-5-2-1-2-1-4	3-3-3-3-4-4-4-4-2-2-2-2-5-5-5-5-1-1-1-1
8	Local_Best	2632	80	6593	5-5-3-5-1-1-2-4-4-2-1-4-5-1-3-2-3-2-3-4	5-5-5-5-1-1-1-1-4-4-4-4-2-2-2-2-3-3-3-3
9	Local_Best	1866	80	7546	3-1-5-1-1-5-2-3-1-5-2-3-2-2-3-4-4-4-5-4	1-1-1-1-5-5-5-5-2-2-2-2-3-3-3-3-4-4-4-4
8	Local_Best	2650	80	734	4-1-3-4-5-1-5-2-1-3-3-5-1-5-2-2-2-4-3-4	1-1-1-1-5-5-5-5-3-3-3-3-2-2-2-2-4-4-4-4
9	Local_Best	2676	80	6377	1-3-2-2-4-1-3-5-5-1-2-5-4-3-1-2-4-5-4-3	2-2-2-2-1-1-1-1-5-5-5-5-3-3-3-3-4-4-4-4

9	Local_Best	2230	80	6568	5-5-1-5-3-4-3-2-1-5-2-4-2-4-3-2-4-3-1-1	5-5-5-5-3-3-3-3-4-4-4-4-2-2-2-2-1-1-1-1
9	Local_Best	2652	80	63	1-5-1-2-3-5-5-3-4-3-2-1-4-2-3-5-1-2-4-4	5-5-5-5-3-3-3-3-1-1-1-1-2-2-2-2-4-4-4-4
10	Local_Best	1476	80	6694	1-2-4-3-2-1-3-1-3-1-5-3-4-2-2-5-4-5-5-4	2-2-2-2-1-1-1-1-3-3-3-3-5-5-5-5-4-4-4-4

trade.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
9	Local_Best	2943	677	9988	4-3-1-1-5-3-3-5-2-4-5-1-2-4-4-2-5-3-2-1	1-1-1-1-4-4-5-5-4-4-5-5-3-3-2-2-3-3-2-2
7	Local_Best	2077	677	4942	4-1-3-2-5-1-3-3-2-5-5-3-5-1-4-4-2-1-2-4	2-2-2-2-5-5-3-3-5-5-3-3-1-1-4-4-1-1-4-4
7	Local_Best	2729	705	5178	3-1-5-5-1-5-3-3-1-4-1-2-5-4-3-4-2-2-2-4	1-1-5-5-1-1-3-3-5-5-3-3-2-2-4-4-2-2-4-4
8	Local_Best	1805	677	5669	5-4-3-1-3-3-5-3-1-4-1-5-5-4-2-2-1-4-2-2	3-3-3-3-1-1-5-5-1-1-5-5-4-4-2-2-4-4-2-2
7	Local_Best	2257	797	4967	3-3-1-2-5-3-2-5-4-4-4-2-5-4-1-1-3-1-5-2	3-3-2-2-3-3-2-2-4-4-1-1-4-4-1-1-5-5-5-5
7	Local_Best	2169	797	10125	1-1-2-4-1-2-4-3-3-3-5-3-1-2-5-5-4-5-4-2	1-1-3-3-1-1-3-3-2-2-5-5-2-2-5-5-4-4-4-4
8	Local_Best	2889	737	25524	4-2-5-3-3-2-4-5-1-1-1-2-5-2-1-4-3-3-5-4	2-2-5-5-2-2-5-5-1-1-1-1-3-3-4-4-3-3-4-4
7	Local_Best	2595	765	5597	3-3-2-2-4-3-1-3-1-1-5-2-1-4-4-4-5-2-5-5	3-3-2-2-3-3-2-2-1-1-4-4-1-1-5-5-4-4-5-5

10	Local_Best	2703	765	6607	5-2-3-1-1-2-4-4-3-1-2-1-2-3-5-4-3-5-5-4	2-2-1-1-2-2-1-1-3-3-4-4-3-3-5-5-4-4-5-5
10	Local_Best	3021	705	6846	4-2-2-5-3-3-5-3-1-4-1-1-2-4-4-3-5-5-1-2	4-4-3-3-4-4-1-1-3-3-1-1-5-5-2-2-5-5-2-2

trap.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
11	Local_Best	7600	2560	13329	6-6-1-2-6-2-4-1-3-5-3-6-2-4-4-3-5-5-3-2-4-5-1-1	6-2-6-2-6-2-6-2-3-3-3-3-4-5-4-5-5-5-4-1-4-1-1-1
11	Local_Best	7840	2560	11197	6-1-4-3-1-3-5-3-2-5-2-2-1-3-5-4-6-4-6-6-4-2-5-1	6-3-3-3-6-3-6-2-6-2-2-2-4-4-5-4-5-4-5-1-5-1-1-1
12	Local_Best	7720	4400	12368	4-2-6-1-1-2-3-6-3-5-6-3-5-2-4-3-6-1-5-2-1-4-4-5	6-2-6-2-2-2-6-5-6-5-5-5-3-3-3-3-1-1-1-1-4-4-4-4
14	Local_Best	5760	2560	14335	6-2-4-1-4-1-6-2-4-5-5-6-3-3-1-2-1-6-4-5-2-5-3-3	4-2-4-2-4-2-4-2-6-6-1-6-1-6-1-5-1-5-5-5-3-3-3-3
11	Local_Best	7400	720	12919	6-5-1-6-2-4-5-6-6-1-3-2-2-4-3-1-2-1-3-4-5-4-5-3	3-6-3-6-3-6-3-6-2-1-2-1-2-1-2-1-5-4-5-4-5-4-5-4
9	Local_Best	6000	2560	9589	3-5-4-1-3-3-2-4-6-4-2-5-2-6-1-6-2-1-5-4-5-3-6-1	3-3-3-3-4-4-2-4-2-4-2-6-2-6-6-6-5-1-5-1-5-1-5-1
12	Local_Best	7720	720	11962	6-3-2-2-4-4-2-4-5-5-2-1-5-6-6-3-3-1-1-5-3-1-6-4	2-4-2-4-2-4-2-4-6-3-6-3-6-3-6-3-1-5-1-5-1-5-1-5
14	Local_Best	7880	4400	13758	1-6-3-5-6-3-2-2-1-4-3-4-2-1-5-6-3-5-6-5-4-2-1-4	1-1-1-1-2-2-2-2-6-6-6-6-3-5-5-5-3-5-3-4-3-4-4-4
10	Local_Best	7720	2560	9998	5-2-4-6-3-2-1-4-5-1-1-3-6-3-3-6-2-6-1-4-5-5-2-4	5-2-5-2-2-2-5-3-5-3-3-3-1-6-1-6-6-6-1-4-1-4-4-4

8	Local_Best	6000	2560	7893	2-6-1-1-6-5-5-4-2-1-2-6-4-3-5-3-5-3-4-1-4-3-2-6	2-1-1-1-2-1-2-5-2-5-5-5-4-3-3-3-4-3-4-6-4-6-6-6

Stochastic HC

basic.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
14	Local_Best	3046	80	12295	2-3-3-2-2-5-3-1-1-1-4-4-3-4-5-5-5-4-1-2	3-3-3-3-2-2-2-2-5-5-5-5-4-4-4-4-1-1-1-1
10	Local_Best	2050	80	7247	4-3-5-4-4-5-4-5-1-2-1-2-3-3-5-1-2-3-2-1	4-4-4-4-5-5-5-5-3-3-3-3-1-1-1-1-2-2-2-2
13	Local_Best	1818	80	9654	3-1-4-3-3-2-4-2-5-3-4-4-5-1-1-2-2-1-5-5	3-3-3-3-2-2-2-2-4-4-4-4-1-1-1-1-5-5-5-5
16	Local_Best	2634	80	11677	5-4-5-1-4-2-4-3-1-3-4-2-5-1-2-3-5-3-2-1	4-4-4-4-1-1-1-1-5-5-5-5-3-3-3-3-2-2-2-2
13	Local_Best	3040	80	11553	3-2-4-4-5-1-3-2-3-5-5-4-1-4-1-2-3-5-2-1	4-4-4-4-2-2-2-2-3-3-3-3-5-5-5-5-1-1-1-1
17	Local_Best	3060	80	14908	2-5-1-2-3-3-4-1-1-5-4-3-5-4-2-4-2-5-3-1	3-3-3-3-4-4-4-4-2-2-2-2-5-5-5-5-1-1-1-1
15	Local_Best	2264	80	10346	1-3-2-1-4-5-3-5-2-5-1-1-4-3-5-2-4-2-3-4	1-1-1-1-3-3-3-3-5-5-5-5-2-2-2-2-4-4-4-4
20	Local_Best	2230	80	12697	3-4-5-3-2-1-2-4-2-5-3-2-1-1-4-3-1-4-5-5	2-2-2-2-3-3-3-3-1-1-1-1-4-4-4-4-5-5-5-5
10	Local_Best	1446	80	6521	3-3-5-3-5-3-5-5-2-1-2-4-1-1-4-4-2-2-4-1	5-5-5-5-3-3-3-3-4-4-4-4-2-2-2-2-1-1-1-1
17	Local_Best	2218	80	1109	2-4-5-1-3-1-3-3-4-3-4-5-1-1-2-2-2-4-5-5	3-3-3-3-4-4-4-4-2-2-2-2-1-1-1-1-5-5-5-5

trade.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
12	Local_Best	3123	797	9758	2-1-3-5-2-1-3-4-5-5-4-3-3-1-4-5-4-2-2-1	1-1-3-3-1-1-3-3-2-2-5-5-2-2-5-5-4-4-4-4
15	Local_Best	1997	677	10096	3-1-4-1-3-2-2-5-4-2-1-1-2-3-4-5-5-3-4-5	1-1-1-1-2-2-4-4-2-2-4-4-3-3-5-5-3-3-5-5
18	Local_Best	2869	677	12459	3-3-1-5-2-4-3-2-2-5-3-5-4-1-5-1-2-4-1-4	2-2-2-2-1-1-5-5-1-1-5-5-3-3-4-4-3-3-4-4
15	Local_Best	2777	737	10641	2-4-4-2-5-4-1-1-3-3-4-3-1-1-5-2-3-2-5-5	4-4-1-1-4-4-1-1-2-2-2-2-3-3-5-5-3-3-5-5
16	Local_Best	2771	765	11487	5-3-2-1-5-2-5-2-2-1-4-1-5-3-3-4-4-3-4-1	5-5-1-1-5-5-1-1-3-3-2-2-3-3-4-4-2-2-4-4
18	Local_Best	2809	765	12908	2-1-2-3-3-1-4-1-5-5-5-4-2-1-3-3-2-4-5-4	2-2-1-1-2-2-1-1-4-4-3-3-4-4-5-5-3-3-5-5
13	Local_Best	2815	765	10478	1-2-4-4-5-3-4-5-5-2-4-2-3-5-2-1-1-3-3-1	2-2-4-4-2-2-4-4-5-5-3-3-5-5-1-1-3-3-1-1
19	Local_Best	2169	797	1225	1-1-2-2-1-4-5-5-4-2-2-4-5-3-3-5-3-1-3-4	5-5-2-2-5-5-2-2-1-1-4-4-1-1-4-4-3-3-3-3
15	Local_Best	2943	1449	10077	4-3-2-4-5-1-3-4-4-1-5-2-2-3-1-3-2-5-5-1	4-4-4-4-3-3-3-3-2-2-5-5-2-2-1-1-5-5-1-1
20	Local_Best	2841	765	13426	5-3-2-3-1-5-2-2-4-4-2-5-4-1-1-5-3-3-4-1	4-4-5-5-4-4-5-5-3-3-1-1-3-3-2-2-1-1-2-2

trap.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
16	Local_Best	7720	2560	28379	6-2-5-4-1-1-5-4-3-2-3-5-1-2-2-4-6-1-5-6-3-3-6-4	2-2-2-2-5-1-5-1-5-1-5-1-4-4-6-4-6-4-6-3-6-3-3-3
15	Local_Best	5720	2560	15924	5-1-4-5-4-2-6-3-4-2-5-3-1-5-6-3-4-2-6-6-1-2-3-1	2-2-4-2-4-2-4-5-4-5-5-5-6-3-6-3-3-3-6-1-6-1-1-1
28	Local_Best	7920	2560	27148	2-6-3-3-4-1-2-6-5-6-4-3-5-1-4-1-3-2-1-5-5-4-6-2	2-2-4-2-4-2-4-1-4-1-1-1-3-3-5-3-5-3-5-6-5-6-6-6
16	Local_Best	7320	2560	16205	4-5-1-5-5-1-5-2-6-2-4-2-6-1-3-4-3-1-4-6-6-2-3-3	5-5-5-5-4-2-4-2-4-2-4-2-3-3-3-3-1-6-1-6-1-6-1-6
24	Local_Best	7760	4400	26371	1-5-4-3-6-4-3-1-5-2-2-2-5-6-4-4-6-5-1-1-2-3-3-6	4-4-4-4-1-5-5-5-1-5-1-2-1-2-2-2-6-6-6-6-3-3-3-3
20	Local_Best	5640	2560	21524	3-1-2-4-5-6-5-6-6-4-5-2-1-6-4-1-3-4-5-2-2-1-3-3	3-6-6-6-3-6-3-2-3-2-2-2-5-4-5-4-5-4-5-4-1-1-1-1
27	Local_Best	7760	4400	26595	3-4-2-6-6-5-2-4-1-3-5-2-2-3-4-1-6-5-6-1-3-1-5-4	3-3-3-3-1-1-1-1-6-2-6-2-6-2-6-2-5-5-5-5-4-4-4-4
20	Local_Best	5880	2560	20885	5-4-2-3-6-5-5-1-4-1-3-4-2-5-2-1-6-3-6-2-4-3-6-1	2-4-2-4-2-4-2-4-1-1-1-1-5-5-5-5-6-3-6-3-6-3-6-3
19	Local_Best	7880	2560	19755	4-3-1-5-2-1-2-3-6-2-3-1-4-6-6-5-1-5-2-5-4-3-6-4	3-3-1-3-1-3-1-5-1-5-2-5-2-5-2-6-2-6-6-6-4-4-4-4
20	Local_Best	7680	2560	21218	2-1-4-5-6-4-1-5-3-5-3-6-5-2-1-3-2-2-4-4-6-3-6-1	6-5-6-5-6-5-6-5-1-4-1-4-1-4-1-4-2-2-2-2-3-3-3-3

Random Restart HC (5)

basic.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State

9	Local_Best	3060	80	7168	4-2-1-4-1-2-3-5-4-2-1-5-5-3-3-1-5-3-4-2	4-4-4-4-1-1-1-1-5-5-5-5-3-3-3-3-2-2-2-2
7	Local_Best	2652	80	507	1-2-2-4-3-3-4-5-5-1-3-5-3-2-4-5-1-4-2-1	2-2-2-2-3-3-3-3-5-5-5-5-4-4-4-4-1-1-1-1
6	Local_Best	3046	80	4256	3-1-1-3-4-4-4-1-5-2-2-2-2-5-5-3-5-4-3-1	1-1-1-1-4-4-4-4-2-2-2-2-5-5-5-5-3-3-3-3
8	Local_Best	2676	80	603	3-2-4-2-5-5-4-4-2-3-1-3-1-2-3-1-5-5-1-4	2-2-2-2-4-4-4-4-3-3-3-3-1-1-1-1-5-5-5-5
10	Local_Best	2666	80	7355	2-2-4-5-1-3-3-4-1-4-5-3-5-4-2-5-1-1-2-3	4-4-4-4-3-3-3-3-1-1-1-1-5-5-5-5-2-2-2-2
10	Local_Best	2278	80	6824	3-1-3-2-1-4-4-5-1-3-5-3-5-4-2-4-2-2-5-1	4-4-4-4-1-1-1-1-3-3-3-3-2-2-2-2-5-5-5-5
9	Local_Best	2286	80	6541	2-3-4-5-2-1-4-3-5-4-5-3-4-2-3-5-1-1-1-2	2-2-2-2-5-5-5-5-4-4-4-4-3-3-3-3-1-1-1-1
7	Local_Best	2658	80	5082	5-2-1-2-3-2-1-5-4-4-1-5-3-1-3-3-4-2-5-4	2-2-2-2-1-1-1-1-5-5-5-5-3-3-3-3-4-4-4-4
8	Local_Best	2638	80	4996	2-2-1-4-2-4-1-4-4-3-5-3-1-5-3-5-3-2-1-5	2-2-2-2-4-4-4-4-5-5-5-5-3-3-3-3-1-1-1-1
9	Local_Best	3060	80	6292	3-4-5-2-5-1-1-2-3-4-1-2-3-4-5-1-3-2-4-5	2-2-2-2-1-1-1-1-5-5-5-5-3-3-3-3-4-4-4-4

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
9	Local_Best	2203	705	617	3-1-4-3-5-4-5-1-2-3-3-1-2-2-4-5-2-5-4-1	5-5-1-1-5-5-3-3-1-1-3-3-2-2-4-4-2-2-4-4

8	Local_Best	2757	677	5777	5-1-5-5-2-3-2-4-2-5-1-4-3-4-3-1-3-2-4-1	5-5-5-5-2-2-4-4-2-2-4-4-3-3-1-1-3-3-1-1
9	Local_Best	2975	705	5854	2-2-1-4-3-1-2-4-5-3-3-5-5-2-4-3-1-5-1-4	2-2-1-1-2-2-3-3-1-1-3-3-5-5-4-4-5-5-4-4
8	Local_Best	2269	677	5559	4-2-3-5-4-5-1-3-3-1-5-1-2-2-4-3-2-5-4-1	5-5-5-5-3-3-1-1-3-3-1-1-2-2-4-4-2-2-4-4
6	Local_Best	2847	677	4239	3-4-5-3-2-4-5-5-2-3-3-5-4-1-4-2-1-1-1-2	3-3-3-3-4-4-5-5-4-4-5-5-1-1-2-2-1-1-2-2
10	Local_Best	2977	705	26844	4-4-5-3-1-2-5-2-2-1-2-4-4-3-1-3-3-5-5-1	4-4-3-3-4-4-2-2-3-3-2-2-5-5-1-1-5-5-1-1
10	Local_Best	3017	705	27477	2-2-1-5-4-4-3-4-3-5-5-3-1-4-1-2-2-5-1-3	2-2-1-1-2-2-5-5-1-1-5-5-4-4-3-3-4-4-3-3
7	Local_Best	3043	677	18146	1-5-5-2-1-2-4-4-5-4-5-1-3-3-2-3-1-3-2-4	5-5-5-5-1-1-4-4-1-1-4-4-3-3-2-2-3-3-2-2
7	Local_Best	2709	705	5323	1-5-2-3-1-3-1-2-5-2-4-1-2-3-4-4-5-3-5-4	5-5-2-2-5-5-1-1-2-2-1-1-3-3-4-4-3-3-4-4
9	Local_Best	2809	677	6173	1-3-3-1-5-3-2-2-5-5-4-4-2-2-3-4-1-1-4-5	2-2-2-2-5-5-3-3-5-5-3-3-1-1-4-4-1-1-4-4

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
12	Local_Best	7880	720	15878	6-5-5-2-4-6-2-4-3-5-2-4-1-3-1-1-1-3-2-4-6-5-6-3	4-2-4-2-4-2-4-2-1-3-1-3-1-3-1-3-6-5-6-5-6-5-6-5
10	Local_Best	5560	2560	14687	2-3-6-5-4-4-3-4-2-4-6-1-5-5-1-6-2-6-3-5-2-1-1-3	2-4-2-4-4-4-2-5-2-5-5-5-3-6-3-6-6-6-3-1-3-1-1-1

12	Local_Best	7400	720	12626	3-4-6-2-5-3-2-1-5-2-6-5-5-6-6-2-1-4-4-3-1-1-3-4	5-2-5-2-5-2-5-2-6-4-6-4-6-4-6-4-1-3-1-3-1-3-1-3
10	Local_Best	7400	2560	1047	1-4-6-1-2-3-1-2-1-2-5-5-4-6-6-3-2-5-3-3-5-6-4-4	1-2-1-2-2-2-1-5-1-5-4-5-4-5-4-3-4-3-3-3-6-6-6-6
10	Local_Best	7600	2560	1066	1-4-2-6-4-5-4-1-4-2-3-5-3-6-5-3-1-2-3-6-5-1-6-2	4-1-1-1-4-1-4-5-4-5-5-5-3-6-3-6-6-6-3-2-3-2-2-2
9	Local_Best	7840	2560	10348	4-3-1-1-1-4-5-2-3-4-3-6-5-1-2-6-5-2-6-4-3-2-6-5	1-1-1-1-4-4-5-4-5-4-5-6-5-6-6-6-3-2-3-2-3-2-3-2
12	Local_Best	5600	720	12381	2-3-4-6-1-4-5-5-6-3-2-1-2-4-2-5-4-1-5-1-6-6-3-3	5-4-5-4-5-4-5-4-2-1-2-1-2-1-2-1-6-3-6-3-6-3-6-3
11	Local_Best	7680	2560	12698	1-1-4-6-3-3-5-6-4-6-4-6-5-3-2-2-1-5-1-3-2-4-5-2	4-6-4-6-4-6-4-6-1-3-1-3-3-3-1-2-1-2-2-2-5-5-5-5
12	Local_Best	7680	720	13734	2-5-4-2-3-1-2-6-6-1-5-6-6-5-4-3-3-5-4-4-1-3-1-2	4-5-4-5-4-5-4-5-6-3-6-3-6-3-6-3-1-2-1-2-1-2-1-2
11	Local_Best	7760	720	12798	6-6-5-4-3-5-5-2-4-1-1-3-3-3-2-1-6-2-1-6-4-2-4-5	5-1-5-1-5-1-5-1-6-3-6-3-6-3-6-3-4-2-4-2-4-2-4-2

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
10	Local_Best	3040	80	10402	4-3-2-3-5-1-1-2-5-4-5-4-3-5-1-2-1-4-2-3	2-2-2-2-5-5-5-5-4-4-4-4-1-1-1-1-3-3-3-3
8	Local_Best	1854	80	582	4-3-1-1-1-5-3-4-4-1-2-5-3-3-5-5-4-2-2-2	1-1-1-1-4-4-4-4-3-3-3-3-5-5-5-5-2-2-2-2
9	Local_Best	1644	80	68	5-4-3-4-4-2-4-1-1-1-2-2-3-1-5-2-3-5-5-3	4-4-4-4-2-2-2-2-1-1-1-1-5-5-5-5-3-3-3-3

8	Local_Best	2676	80	5639	2-3-4-4-1-3-2-1-1-5-5-2-4-5-4-5-3-1-3-2	2-2-2-2-1-1-1-1-5-5-5-5-4-4-4-4-3-3-3-3
7	Local_Best	2268	80	4793	1-5-3-1-1-3-4-3-2-4-4-5-3-4-5-5-2-2-2-1	1-1-1-1-3-3-3-3-4-4-4-4-5-5-5-5-2-2-2-2
9	Local_Best	2240	80	5889	5-4-1-2-4-3-4-1-3-5-4-2-5-1-2-5-3-1-2-3	2-2-2-2-4-4-4-4-1-1-1-1-5-5-5-5-3-3-3-3
9	Local_Best	2634	80	6366	3-4-4-3-3-4-5-1-1-2-3-2-2-5-5-1-5-4-1-2	4-4-4-4-3-3-3-3-2-2-2-2-5-5-5-5-1-1-1-1
7	Local_Best	1846	80	4768	3-1-5-3-3-4-5-2-5-4-1-3-1-4-4-1-2-5-2-2	3-3-3-3-5-5-5-5-4-4-4-4-1-1-1-1-2-2-2-2
8	Local_Best	2238	80	557	1-4-3-3-4-2-1-5-3-4-1-4-5-1-2-2-5-3-2-5	3-3-3-3-1-1-1-1-4-4-4-4-2-2-2-2-5-5-5-5
9	Local_Best	2608	80	5993	1-3-5-4-4-4-2-3-2-5-1-5-4-5-1-2-2-1-3-3	2-2-2-2-4-4-4-4-5-5-5-5-1-1-1-1-3-3-3-3

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
6	Local_Best	2623	677	4135	1-5-1-1-1-5-3-4-5-4-2-4-2-2-4-5-2-3-3-3	1-1-1-1-5-5-4-4-5-5-4-4-2-2-3-3-2-2-3-3
8	Local_Best	2183	677	5705	5-2-2-5-1-4-3-5-4-3-3-1-1-5-2-3-4-4-2-1	5-5-5-5-1-1-3-3-1-1-3-3-4-4-2-2-4-4-2-2
9	Local_Best	3101	677	6489	3-4-5-5-2-3-3-4-2-5-4-1-1-1-3-1-2-4-5-2	3-3-3-3-2-2-4-4-2-2-4-4-1-1-5-5-1-1-5-5
8	Local_Best	2849	677	5757	4-5-2-5-1-5-1-2-3-4-3-1-3-3-1-2-4-5-4-2	5-5-5-5-4-4-1-1-4-4-1-1-3-3-2-2-3-3-2-2

7	Local_Best	1865	677	5265	2-5-1-5-3-1-2-1-5-5-4-4-1-3-2-4-3-3-4-2	1-1-1-1-5-5-2-2-5-5-2-2-3-3-4-4-3-3-4-4
6	Local_Best	2263	677	4177	3-5-5-5-2-4-4-1-3-2-1-1-4-1-3-4-2-2-3-5	5-5-5-5-4-4-1-1-4-4-1-1-2-2-3-3-2-2-3-3
10	Local_Best	2789	677	6669	5-5-3-1-3-4-4-5-4-4-2-2-1-2-2-1-3-5-1-3	5-5-5-5-4-4-2-2-4-4-2-2-3-3-1-1-3-3-1-1
10	Local_Best	2843	677	713	1-5-2-2-4-1-1-5-4-5-3-2-3-5-2-4-1-3-3-4	1-1-1-1-3-3-2-2-3-3-2-2-5-5-4-4-5-5-4-4
9	Local_Best	3057	677	6501	5-5-4-2-4-3-2-3-1-3-2-4-1-2-5-1-4-5-3-1	2-2-2-2-3-3-4-4-3-3-4-4-5-5-1-1-5-5-1-1
9	Local_Best	2251	677	6273	2-2-5-5-4-3-4-1-3-3-4-5-4-1-1-2-3-2-1-5	5-5-5-5-3-3-4-4-3-3-4-4-2-2-1-1-2-2-1-1

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
12	Local_Best	5720	720	1332	3-5-1-5-2-4-2-4-4-5-1-6-5-3-2-3-4-6-6-1-3-6-2-1	2-4-2-4-2-4-2-4-5-3-5-3-5-3-5-3-6-1-6-1-6-1-6-1
11	Local_Best	5760	720	11102	6-5-3-5-3-2-2-1-1-4-3-4-1-6-4-6-1-6-3-4-5-2-5-2	3-4-3-4-3-4-3-4-1-6-1-6-1-6-1-6-5-2-5-2-5-2-5-2
14	Local_Best	7960	2560	14135	4-5-6-3-1-6-6-2-4-1-3-5-2-2-1-4-1-5-3-3-4-5-2-6	6-6-1-6-1-6-1-2-1-2-2-2-3-3-4-3-4-3-4-5-4-5-5-5
11	Local_Best	7480	2560	11509	5-6-3-4-6-6-1-4-2-6-4-5-5-2-1-5-3-1-2-1-3-3-2-4	6-4-6-4-6-4-6-4-3-5-3-5-5-5-3-1-3-1-1-1-2-2-2-2
14	Local_Best	7960	720	16369	4-2-4-4-1-2-6-6-3-1-5-1-2-6-5-5-3-4-1-3-2-6-5-3	4-6-4-6-4-6-4-6-2-1-2-1-2-1-2-1-5-3-5-3-5-3-5-3

12	Local_Best	5640	720	1271	4-1-3-3-4-2-5-2-5-5-3-3-1-6-6-2-2-1-6-6-4-5-4-1	5-3-5-3-5-3-5-3-4-2-4-2-4-2-4-2-6-1-6-1-6-1-6-1
9	Local_Best	5520	720	11349	6-1-2-4-4-3-2-6-4-4-5-6-2-1-2-1-3-5-3-5-1-5-3-6	4-6-4-6-4-6-4-6-2-1-2-1-2-1-2-1-3-5-3-5-3-5-3-5
9	Local_Best	7440	720	8735	2-6-3-2-3-1-2-1-3-4-5-1-2-4-1-5-6-6-6-3-4-5-5-4	3-1-3-1-3-1-3-1-2-4-2-4-2-4-2-4-6-5-6-5-6-5-6-5
10	Local_Best	7480	2560	1015	4-6-6-6-3-5-1-4-2-3-4-2-6-4-5-5-1-2-5-1-3-3-2-1	6-6-6-6-2-4-2-4-2-4-2-4-3-5-5-5-3-5-3-1-3-1-1-1
11	Local_Best	7560	720	14732	1-2-4-3-6-6-4-5-6-4-4-2-5-3-3-1-1-1-2-3-6-5-2-5	4-2-4-2-4-2-4-2-6-1-6-1-6-1-6-1-3-5-3-5-3-5-3-5

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
9	Local_Best	3060	80	8341	2-1-3-4-3-3-5-5-1-1-4-4-5-2-2-3-2-1-5-4	3-3-3-3-4-4-4-4-1-1-1-1-2-2-2-2-5-5-5-5
9	Local_Best	2266	80	6612	2-4-3-1-2-1-3-2-4-3-1-4-2-1-4-5-5-5-3-5	2-2-2-2-1-1-1-1-3-3-3-3-4-4-4-4-5-5-5-5
9	Local_Best	2224	80	22828	2-5-5-4-1-3-3-4-4-2-4-1-1-5-3-5-1-3-2-2	1-1-1-1-4-4-4-4-3-3-3-3-5-5-5-5-2-2-2-2
9	Local_Best	3072	80	7823	4-5-5-2-3-5-2-3-4-1-3-1-1-4-2-2-1-5-3-4	5-5-5-5-3-3-3-3-1-1-1-1-2-2-2-2-4-4-4-4
9	Local_Best	2670	80	6056	4-4-3-2-2-5-3-2-4-1-5-3-4-1-5-3-5-1-1-2	3-3-3-3-2-2-2-2-4-4-4-4-1-1-1-1-5-5-5-5
11	Local_Best	3060	80	7362	1-2-4-5-4-5-5-1-3-1-3-2-4-3-2-4-1-5-2-3	5-5-5-5-2-2-2-2-1-1-1-1-4-4-4-4-3-3-3-3

10	Local_Best	2242	80	6687	4-2-3-5-3-4-1-5-1-2-5-5-4-3-2-2-3-4-1-1	3-3-3-3-5-5-5-5-4-4-4-4-2-2-2-2-1-1-1-1
10	Local_Best	2682	80	6809	3-4-5-4-2-3-5-5-3-4-2-1-3-2-5-1-1-1-2-4	5-5-5-5-4-4-4-4-3-3-3-3-1-1-1-1-2-2-2-2
8	Local_Best	2656	80	5178	1-1-5-3-5-2-2-1-4-4-4-2-3-1-2-5-3-4-3-5	1-1-1-1-2-2-2-2-4-4-4-4-5-5-5-5-3-3-3-3
9	Local_Best	2264	80	6054	1-4-5-3-1-1-4-3-5-2-3-1-2-5-5-2-4-3-4-2	3-3-3-3-1-1-1-1-5-5-5-5-2-2-2-2-4-4-4-4

trade.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
7	Local_Best	2929	677	6823	4-5-3-5-4-5-2-1-1-1-2-2-3-4-5-4-1-2-3-3	5-5-5-5-4-4-2-2-4-4-2-2-1-1-3-3-1-1-3-3
8	Local_Best	3057	677	5877	5-3-3-4-1-2-5-2-1-1-4-3-4-4-2-5-3-2-1-5	4-4-4-4-1-1-2-2-1-1-2-2-3-3-5-5-3-3-5-5
11	Local_Best	3035	677	8297	1-4-3-1-2-5-5-4-3-1-1-5-4-3-4-2-2-2-3-5	3-3-3-3-1-1-4-4-1-1-4-4-2-2-5-5-2-2-5-5
10	Local_Best	3003	677	6683	5-1-3-1-3-5-5-2-4-4-4-1-3-4-5-2-1-2-2-3	5-5-5-5-3-3-1-1-3-3-1-1-4-4-2-2-4-4-2-2
8	Local_Best	2815	677	5809	1-3-3-3-5-4-2-3-2-1-4-1-5-2-5-2-5-1-4-4	3-3-3-3-1-1-2-2-1-1-2-2-5-5-4-4-5-5-4-4
9	Local_Best	3055	677	5815	2-4-1-1-3-1-5-2-4-2-4-3-5-5-1-5-2-3-3-4	1-1-1-1-2-2-4-4-2-2-4-4-5-5-3-3-5-5-3-3
8	Local_Best	2015	677	5542	3-5-4-3-5-3-1-3-1-5-2-4-4-2-4-1-5-2-1-2	3-3-3-3-5-5-4-4-5-5-4-4-2-2-1-1-2-2-1-1

9	Local_Best	2983	677	6093	3-1-5-2-4-1-4-2-5-5-3-3-2-1-3-4-4-5-1-2	4-4-4-4-1-1-3-3-1-1-3-3-5-5-2-2-5-5-2-2
12	Local_Best	2637	677	8605	2-3-3-1-1-5-4-5-4-3-4-2-1-1-4-2-2-3-5-5	4-4-4-4-1-1-2-2-1-1-2-2-3-3-5-5-3-3-5-5
9	Local_Best	2971	677	6134	5-4-2-1-2-1-3-1-3-5-5-4-5-3-2-4-3-4-2-1	1-1-1-1-5-5-4-4-5-5-4-4-3-3-2-2-3-3-2-2

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
9	Local_Best	7720	720	9395	6-6-4-1-2-2-6-4-5-3-3-2-1-3-4-6-2-5-1-4-3-5-1-5	6-2-6-2-6-2-6-2-4-3-4-3-4-3-4-3-1-5-1-5-1-5-1-5
11	Local_Best	5560	720	10965	2-2-5-5-3-2-6-4-6-1-5-2-4-3-6-3-4-5-4-3-1-1-6-1	5-2-5-2-5-2-5-2-4-3-4-3-4-3-4-3-6-1-6-1-6-1-6-1
10	Local_Best	7640	720	11289	1-2-5-3-5-6-4-3-6-3-4-2-1-4-3-2-4-1-6-1-6-2-5-5	5-3-5-3-5-3-5-3-4-2-4-2-4-2-4-2-6-1-6-1-6-1-6-1
9	Local_Best	6000	2560	6352	6-4-5-5-2-5-1-6-1-3-1-2-4-2-3-4-1-6-3-4-3-6-2-5	5-5-5-5-4-4-1-4-1-4-1-2-1-2-2-2-3-6-3-6-3-6-3-6
11	Local_Best	5720	720	575	6-1-4-5-4-2-3-1-6-1-2-5-1-3-3-2-6-5-6-2-5-4-3-4	2-1-2-1-2-1-2-1-6-5-6-5-6-5-6-5-3-4-3-4-3-4-3-4
15	Local_Best	7400	720	8438	2-1-6-4-3-5-1-1-1-6-6-3-3-2-4-5-3-6-2-4-5-4-2-5	1-6-1-6-1-6-1-6-3-5-3-5-3-5-3-5-2-4-2-4-2-4-2-4
14	Local_Best	7600	720	8018	5-3-1-4-3-3-5-2-6-5-6-1-1-1-6-4-5-2-2-6-2-3-4-4	5-1-5-1-5-1-5-1-6-2-6-2-6-2-6-2-3-4-3-4-3-4-3-4
13	Local_Best	8000	720	6637	3-3-4-2-6-6-1-5-2-5-4-4-1-6-2-3-1-6-5-2-4-5-3-1	4-5-4-5-4-5-4-5-1-6-1-6-1-6-1-6-2-3-2-3-2-3-2-3
12	Local_Best	7760	720	21691	1-5-3-6-1-4-2-5-2-6-6-3-4-2-1-1-6-3-4-3-5-2-5-4	1-6-1-6-1-6-1-6-2-3-2-3-2-3-2-3-4-5-4-5-4-5-4-5

11	Local_Best	7600	720	6069	4-6-1-2-4-6-5-3-1-2-5-3-6-5-6-5-4-2-3-4-2-3-1-1	4-2-4-2-4-2-4-2-6-5-6-5-6-5-6-5-3-1-3-1-3-1-3-1

Stochastic random restart (5)

basic.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
15	Local_Best	2662	80	10942	1-5-3-3-4-2-5-2-2-1-3-4-2-5-5-1-4-3-4-1	3-3-3-3-2-2-2-2-1-1-1-1-5-5-5-5-4-4-4-4
18	Local_Best	3070	80	15382	4-3-1-4-2-5-1-2-5-4-5-3-2-3-4-1-3-5-2-1	4-4-4-4-5-5-5-5-1-1-1-1-3-3-3-3-2-2-2-2
14	Local_Best	2244	80	9664	3-4-5-2-3-1-5-5-2-2-5-4-2-1-4-3-3-1-1-4	5-5-5-5-2-2-2-2-3-3-3-3-1-1-1-1-4-4-4-4
17	Local_Best	2250	80	11653	3-2-5-5-5-4-1-1-4-3-5-4-1-3-3-2-2-1-2-4	5-5-5-5-4-4-4-4-3-3-3-3-1-1-1-1-2-2-2-2
16	Local_Best	2666	80	10778	1-3-2-2-4-3-3-1-5-2-4-5-4-3-5-4-1-1-5-2	4-4-4-4-1-1-1-1-2-2-2-2-5-5-5-5-3-3-3-3
15	Local_Best	2434	80	10305	4-2-5-2-2-1-2-1-4-1-5-5-3-3-4-3-3-5-1-4	2-2-2-2-5-5-5-5-1-1-1-1-4-4-4-4-3-3-3-3
15	Local_Best	2648	80	9797	1-4-5-5-5-4-4-1-3-5-2-2-3-2-3-1-3-2-1-4	4-4-4-4-5-5-5-5-2-2-2-2-1-1-1-1-3-3-3-3
14	Local_Best	3040	80	10016	5-3-5-2-1-3-5-2-4-3-1-4-2-5-4-1-1-3-4-2	5-5-5-5-3-3-3-3-1-1-1-1-4-4-4-4-2-2-2-2
22	Local_Best	2672	80	15623	1-5-1-4-2-3-2-5-2-3-4-4-5-5-3-4-2-1-3-1	5-5-5-5-3-3-3-3-1-1-1-1-4-4-4-4-2-2-2-2
14	Local_Best	2252	80	1008	1-3-4-2-2-4-4-5-2-1-2-3-4-3-5-5-5-1-3-1	2-2-2-2-4-4-4-4-1-1-1-1-5-5-5-5-3-3-3-3

trade.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
14	Local_Best	2989	677	11915	4-3-1-1-3-2-4-5-5-4-3-5-2-2-1-5-4-3-1-2	3-3-3-3-2-2-5-5-2-2-5-5-4-4-1-1-4-4-1-1
17	Local_Best	3123	677	12264	4-5-3-1-2-3-4-2-3-2-5-5-5-4-2-1-1-1-3-4	1-1-1-1-4-4-3-3-4-4-3-3-5-5-2-2-5-5-2-2
16	Local_Best	2809	677	11013	5-3-5-5-2-3-5-1-3-4-2-4-2-4-3-1-4-1-1-2	5-5-5-5-3-3-2-2-3-3-2-2-4-4-1-1-4-4-1-1
14	Local_Best	2921	677	38385	2-5-1-1-3-5-3-2-1-5-2-3-4-1-5-3-4-4-2-4	3-3-3-3-1-1-2-2-1-1-2-2-4-4-5-5-4-4-5-5
11	Local_Best	2723	705	8369	3-4-1-3-1-3-5-5-1-4-2-5-1-2-4-5-3-2-4-2	3-3-2-2-3-3-4-4-2-2-4-4-1-1-5-5-1-1-5-5
13	Local_Best	3049	677	41365	3-5-4-1-5-5-2-2-1-3-1-4-4-4-3-3-1-2-5-2	4-4-4-4-5-5-3-3-5-5-3-3-1-1-2-2-1-1-2-2
19	Local_Best	2343	677	14511	4-1-3-3-4-4-5-5-2-1-3-2-2-5-5-1-4-1-3-2	5-5-5-5-4-4-3-3-4-4-3-3-2-2-1-1-2-2-1-1
10	Local_Best	2049	677	696	4-3-1-1-3-5-3-2-4-4-1-2-5-5-4-1-5-3-2-2	3-3-3-3-4-4-1-1-4-4-1-1-5-5-2-2-5-5-2-2
20	Local_Best	2831	677	13433	4-1-2-3-5-5-3-5-3-4-2-1-3-5-4-1-4-2-2-1	5-5-5-5-3-3-1-1-3-3-1-1-4-4-2-2-4-4-2-2
13	Local_Best	3103	677	9229	1-3-2-3-2-5-1-4-1-4-2-5-5-4-5-3-3-4-1-2	2-2-2-2-3-3-5-5-3-3-5-5-4-4-1-1-4-4-1-1

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
24	Local_Best	7520	2560	27255	4-6-3-4-2-4-1-3-5-5-6-4-2-5-3-1-5-1-3-2-1-6-6-2	5-5-5-5-3-1-1-1-3-1-3-4-3-4-4-4-6-2-6-2-6-2-6-2
19	Local_Best	5880	2560	57212	5-5-6-5-3-4-4-4-1-2-2-3-6-3-1-4-1-3-6-2-5-2-6-1	2-2-2-2-1-3-1-3-1-3-1-3-6-5-5-5-6-5-6-4-6-4-4-4
17	Local_Best	8040	720	18703	2-3-1-1-6-2-5-5-3-4-2-5-4-1-6-6-3-4-2-4-3-6-1-5	2-5-2-5-2-5-2-5-4-1-4-1-4-1-4-1-3-6-3-6-3-6-3-6
20	Local_Best	4120	2560	22546	3-5-1-2-1-2-4-5-4-6-1-6-6-3-4-3-2-5-6-3-1-2-4-5	4-2-4-2-4-2-4-2-1-6-1-6-1-6-1-6-5-5-5-5-3-3-3-3
17	Local_Best	5720	2560	18713	2-1-3-4-3-2-3-6-5-4-1-1-2-6-1-6-2-4-6-5-5-5-3-4	2-1-1-1-2-1-2-6-2-6-6-6-3-4-3-4-4-4-3-5-3-5-5-5
19	Local_Best	7880	720	89931	1-4-2-3-5-6-6-5-4-6-3-2-2-3-6-1-5-1-4-4-5-3-2-1	2-3-2-3-2-3-2-3-6-4-6-4-6-4-6-4-5-1-5-1-5-1-5-1
22	Local_Best	7720	2560	23505	2-4-3-5-4-3-5-1-3-6-4-5-2-6-1-4-6-5-3-1-6-2-1-2	3-4-4-4-3-4-3-2-3-2-6-2-6-2-6-5-6-5-5-5-1-1-1-1
17	Local_Best	7640	2560	17943	5-3-6-2-5-1-1-6-6-4-4-1-4-1-2-3-6-3-5-5-4-2-3-2	1-1-1-1-6-5-5-5-6-5-6-2-6-2-2-2-4-3-4-3-4-3-4-3
23	Local_Best	7560	2560	22322	2-6-1-5-3-3-2-6-5-4-3-1-6-1-3-2-2-1-4-6-4-5-4-5	3-3-3-3-2-1-2-1-1-1-2-6-2-6-6-6-4-5-4-5-4-5-4-5
19	Local_Best	5840	2560	19718	1-3-6-5-3-2-6-5-2-6-5-3-1-3-1-4-2-4-5-4-4-6-1-2	2-2-2-2-6-5-6-5-5-5-6-4-6-4-1-4-1-4-1-3-1-3-3-3

Stochastic random restart (15)

basic.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State

14	Local_Best	2210	80	9658	3-5-2-4-1-1-1-2-2-2-4-5-4-1-3-3-3-4-5-5	1-1-1-1-2-2-2-2-4-4-4-4-3-3-3-3-5-5-5-5
15	Local_Best	2676	80	11076	2-5-5-4-3-1-1-5-4-4-2-1-3-3-3-4-2-1-5-2	5-5-5-5-2-2-2-2-3-3-3-3-1-1-1-1-4-4-4-4
15	Local_Best	2240	80	977	5-3-4-2-4-3-5-4-2-5-2-4-1-3-3-1-5-2-1-1	3-3-3-3-4-4-4-4-2-2-2-2-5-5-5-5-1-1-1-1
15	Local_Best	2258	80	1136	5-4-4-1-2-5-2-4-3-3-4-3-2-5-1-3-5-1-2-1	4-4-4-4-3-3-3-3-5-5-5-5-1-1-1-1-2-2-2-2
14	Local_Best	3072	80	9287	2-4-3-5-4-1-1-3-3-4-2-2-5-5-1-1-2-5-4-3	4-4-4-4-1-1-1-1-5-5-5-5-3-3-3-3-2-2-2-2
11	Local_Best	2274	80	8565	4-1-2-4-5-3-3-2-5-3-3-1-4-1-2-1-4-5-2-5	4-4-4-4-3-3-3-3-1-1-1-1-5-5-5-5-2-2-2-2
18	Local_Best	3040	80	12899	2-3-3-5-4-2-1-5-5-3-4-2-1-5-4-1-3-4-2-1	5-5-5-5-3-3-3-3-4-4-4-4-1-1-1-1-2-2-2-2
16	Local_Best	1874	80	6709	1-5-1-4-4-3-5-3-3-3-2-4-4-1-1-5-2-2-5-2	3-3-3-3-4-4-4-4-1-1-1-1-5-5-5-5-2-2-2-2
18	Local_Best	2672	80	7346	5-4-1-4-2-5-2-5-3-4-1-1-4-3-2-5-3-3-1-2	4-4-4-4-1-1-1-1-3-3-3-3-2-2-2-2-5-5-5-5
14	Local_Best	1878	80	5354	5-2-1-5-5-4-3-3-2-2-2-1-1-1-5-4-4-3-4-3	1-1-1-1-5-5-5-5-2-2-2-2-4-4-4-4-3-3-3-3

trade.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
17	Local_Best	2197	677	13565	3-4-1-3-5-2-1-3-4-2-5-2-1-4-5-2-1-4-5-3	1-1-1-1-4-4-5-5-4-4-5-5-2-2-3-3-2-2-3-3

19	Local_Best	2949	677	15307	3-1-3-5-2-4-3-1-3-2-1-4-4-2-5-2-5-5-1-4	4-4-4-4-2-2-3-3-2-2-3-3-5-5-1-1-5-5-1-1
14	Local_Best	2893	677	10853	3-2-2-4-1-1-1-4-3-4-1-4-2-3-5-3-5-5-5-2	1-1-1-1-3-3-4-4-3-3-4-4-5-5-2-2-5-5-2-2
17	Local_Best	2817	677	928	4-1-5-5-1-4-3-3-5-5-1-4-4-2-3-2-2-3-2-1	1-1-1-1-5-5-3-3-5-5-3-3-4-4-2-2-4-4-2-2
20	Local_Best	3033	677	8673	3-5-5-3-5-4-4-2-2-1-4-5-2-1-2-4-3-1-3-1	1-1-1-1-2-2-3-3-2-2-3-3-5-5-4-4-5-5-4-4
14	Local_Best	2983	677	604	2-2-4-3-1-4-5-1-2-1-2-5-5-5-3-3-3-4-4-1	1-1-1-1-2-2-3-3-2-2-3-3-5-5-4-4-5-5-4-4
10	Local_Best	2923	677	4349	4-1-3-4-3-4-5-3-1-5-3-2-1-5-2-2-5-2-4-1	4-4-4-4-1-1-3-3-1-1-3-3-5-5-2-2-5-5-2-2
18	Local_Best	3109	677	7607	4-3-4-1-2-1-5-5-5-2-1-4-3-2-3-5-1-4-3-2	5-5-5-5-1-1-3-3-1-1-3-3-4-4-2-2-4-4-2-2
20	Local_Best	3177	705	923	5-2-4-4-2-1-1-4-5-3-3-5-1-5-2-3-2-3-4-1	1-1-5-5-1-1-3-3-5-5-3-3-2-2-4-4-2-2-4-4
13	Local_Best	2569	677	5645	2-4-4-1-5-2-1-3-1-2-2-1-3-4-3-5-4-3-5-5	2-2-2-2-3-3-1-1-3-3-1-1-4-4-5-5-4-4-5-5

trap.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
26	Local_Best	7960	2560	19021	5-3-1-2-6-4-5-6-2-4-3-1-2-6-4-3-2-1-5-3-1-6-4-5	2-2-5-2-5-2-5-3-5-3-3-3-1-6-1-6-1-6-1-6-4-4-4-4
18	Local_Best	7600	720	15302	5-2-2-3-2-6-1-5-3-5-3-1-6-6-3-4-1-2-1-5-4-6-4-4	5-2-5-2-5-2-5-2-3-6-3-6-3-6-3-6-1-4-1-4-1-4-1-4

22	Local_Best	7680	720	14999	4-5-2-2-6-6-3-4-1-1-3-4-3-4-5-3-2-1-6-5-5-6-1-2	3-4-3-4-3-4-3-4-1-6-1-6-1-6-1-6-2-5-2-5-2-5-2-5
22	Local_Best	7600	720	14586	3-2-1-5-4-5-5-1-6-1-3-4-4-3-5-2-3-2-1-4-6-6-6-2	4-5-4-5-4-5-4-5-3-1-3-1-3-1-3-1-6-2-6-2-6-2-6-2
19	Local_Best	7720	2560	13045	5-3-4-4-4-6-2-5-1-6-3-2-5-1-2-5-1-4-1-6-2-3-6-3	4-5-5-5-4-5-4-1-4-1-1-1-2-6-2-6-2-6-2-6-3-3-3-3
21	Local_Best	3760	720	15273	2-5-2-3-6-4-6-4-1-1-6-3-6-1-3-5-4-4-5-1-2-3-2-5	6-4-6-4-6-4-6-4-2-3-2-3-2-3-2-3-1-5-1-5-1-5-1-5
18	Local_Best	5720	2560	10925	4-5-3-1-2-2-3-6-5-3-1-5-2-6-4-2-1-5-1-3-4-6-4-6	5-5-5-5-1-2-2-2-1-2-1-3-1-3-4-3-4-3-4-6-4-6-6-6
18	Local_Best	7880	2560	12065	1-4-3-4-5-2-3-2-1-6-6-1-5-5-2-6-6-4-4-2-3-3-1-5	2-2-3-2-3-2-3-6-3-6-6-6-1-1-1-1-4-5-4-5-4-5-4-5
19	Local_Best	5640	2560	11991	6-5-2-1-5-3-2-2-5-2-4-1-4-3-3-6-4-1-4-3-6-6-5-1	2-2-2-2-5-3-5-3-5-3-5-3-1-1-1-1-4-6-4-6-4-6-4-6
18	Local_Best	5520	2560	13462	4-3-4-2-3-6-4-2-4-3-6-5-6-1-1-5-6-1-2-1-5-2-3-5	4-6-4-6-4-6-4-6-1-2-1-2-1-2-1-2-5-5-5-5-3-3-3-3

Stochastic random restart (30)

basic.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
20	Local_Best	2262	80	1051	1-5-1-3-2-4-2-4-3-2-3-2-3-1-4-5-4-5-1-5	1-1-1-1-3-3-3-3-4-4-4-4-5-5-5-5-2-2-2-2
17	Local_Best	2632	80	9144	3-5-5-4-5-4-4-1-3-2-1-2-4-5-3-2-1-2-1-3	5-5-5-5-4-4-4-4-3-3-3-3-1-1-1-1-2-2-2-2
18	Local_Best	3060	80	7666	4-3-5-1-4-2-4-1-2-5-3-1-5-3-2-5-1-4-3-2	4-4-4-4-5-5-5-5-2-2-2-2-1-1-1-1-3-3-3-3

15	Local_Best	3040	80	6781	2-1-4-3-5-1-5-5-4-4-2-1-3-1-3-2-5-3-4-2	1-1-1-1-4-4-4-4-5-5-5-5-3-3-3-3-2-2-2-2
13	Local_Best	3048	80	5869	1-5-2-4-4-2-3-5-3-2-1-4-1-1-5-3-4-2-5-3	5-5-5-5-4-4-4-4-1-1-1-1-3-3-3-3-2-2-2-2
21	Local_Best	3046	80	9259	1-5-1-4-5-4-4-3-2-5-3-2-5-3-1-2-1-4-3-2	3-3-3-3-2-2-2-2-1-1-1-1-4-4-4-4-5-5-5-5
18	Local_Best	2632	80	7608	1-5-3-1-4-1-4-2-4-3-2-3-4-3-5-5-2-5-2-1	2-2-2-2-1-1-1-1-3-3-3-3-5-5-5-5-4-4-4-4
20	Local_Best	2622	80	8486	5-1-3-5-2-4-1-2-3-4-1-4-5-4-2-5-1-2-3-3	1-1-1-1-4-4-4-4-2-2-2-2-5-5-5-5-3-3-3-3
16	Local_Best	2672	80	6565	4-1-2-3-5-2-1-4-5-2-3-4-4-1-3-1-5-5-3-2	5-5-5-5-4-4-4-4-1-1-1-1-2-2-2-2-3-3-3-3
15	Local_Best	3040	80	721	5-1-4-4-5-4-3-5-3-3-2-1-1-5-2-2-1-2-3-4	4-4-4-4-3-3-3-3-1-1-1-1-5-5-5-5-2-2-2-2

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
13	Local_Best	2071	677	7802	5-1-4-5-3-5-1-1-5-2-3-4-2-2-3-4-2-4-1-3	5-5-5-5-1-1-4-4-1-1-4-4-2-2-3-3-2-2-3-3
17	Local_Best	2869	677	2145	3-5-4-1-3-3-1-3-5-1-4-2-2-4-2-5-4-2-1-5	3-3-3-3-1-1-4-4-1-1-4-4-5-5-2-2-5-5-2-2
19	Local_Best	3141	677	8167	3-4-4-5-1-1-3-4-5-2-1-2-3-5-4-5-2-2-3-1	1-1-1-1-5-5-3-3-5-5-3-3-4-4-2-2-4-4-2-2
14	Local_Best	2069	677	6858	2-1-4-3-1-4-4-5-1-4-5-2-5-2-3-5-1-3-2-3	4-4-4-4-3-3-5-5-3-3-5-5-1-1-2-2-1-1-2-2

14	Local_Best	2909	677	611	5-5-1-4-3-2-5-5-4-1-4-3-2-2-1-3-3-4-2-1	5-5-5-5-2-2-1-1-2-2-1-1-4-4-3-3-4-4-3-3
14	Local_Best	2911	677	5834	5-3-1-2-4-5-3-4-1-4-5-3-3-2-5-1-1-4-2-2	1-1-1-1-3-3-5-5-3-3-5-5-4-4-2-2-4-4-2-2
20	Local_Best	2909	677	8733	1-5-1-3-5-2-3-4-4-2-2-5-1-4-3-4-1-2-5-3	4-4-4-4-2-2-5-5-2-2-5-5-1-1-3-3-1-1-3-3
8	Local_Best	2801	677	4253	1-4-2-4-1-4-5-4-5-2-2-2-3-3-1-5-1-3-3-5	1-1-1-1-4-4-2-2-4-4-2-2-3-3-5-5-3-3-5-5
15	Local_Best	2005	677	6603	3-5-3-3-2-5-1-1-2-5-4-1-3-1-4-4-2-2-5-4	1-1-1-1-3-3-5-5-3-3-5-5-2-2-4-4-2-2-4-4
16	Local_Best	2983	677	11153	1-4-1-2-3-5-2-3-4-3-5-4-2-1-1-5-4-2-3-5	1-1-1-1-4-4-2-2-4-4-2-2-3-3-5-5-3-3-5-5

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
13	Local_Best	5200	720	3196	4-6-5-5-3-6-3-5-6-6-1-1-5-1-2-1-4-2-4-3-3-2-4-2	1-6-1-6-1-6-1-6-3-5-3-5-3-5-3-5-4-2-4-2-4-2-4-2
21	Local_Best	7840	720	17816	3-4-3-6-2-4-5-1-2-6-3-4-5-2-6-6-5-5-1-1-2-3-4-1	2-6-2-6-2-6-2-6-5-1-5-1-5-1-5-1-4-3-4-3-4-3-4-3
16	Local_Best	7600	720	10205	6-2-5-3-1-4-3-4-3-2-3-4-2-1-5-5-5-6-6-1-1-4-2-6	3-2-3-2-3-2-3-2-1-4-1-4-1-4-1-4-5-6-5-6-5-6-5-6
21	Local_Best	8000	720	1367	5-3-2-6-3-5-2-4-1-1-1-4-5-2-6-3-4-2-6-3-4-5-1-6	3-5-3-5-3-5-3-5-2-6-2-6-2-6-2-6-1-4-1-4-1-4-1-4
16	Local_Best	7320	720	10246	2-1-3-6-2-2-2-5-3-5-5-3-4-4-6-1-5-6-3-1-6-4-1-4	3-5-3-5-3-5-3-5-2-1-2-1-2-1-2-1-4-6-4-6-4-6-4-6

16	Local_Best	7840	720	10258	4-2-3-2-5-3-4-1-3-2-6-1-4-2-5-1-5-6-6-4-1-6-3-5	4-2-4-2-4-2-4-2-3-1-3-1-3-1-3-1-5-6-5-6-5-6-5-6
23	Local_Best	5880	720	1469	6-3-5-3-4-5-4-2-6-5-1-2-1-1-3-5-6-1-4-2-2-6-4-3	1-2-1-2-1-2-1-2-6-3-6-3-6-3-6-3-4-5-4-5-4-5-4-5
21	Local_Best	7920	2560	13465	3-1-3-5-5-5-2-4-6-2-2-4-4-3-1-6-2-1-1-6-4-5-6-3	5-5-5-5-2-1-2-1-2-1-2-1-3-4-4-4-3-4-3-6-3-6-6-6
27	Local_Best	7920	720	27484	6-1-1-2-6-3-4-3-4-5-1-3-5-6-5-2-4-6-1-5-2-3-2-4	6-2-6-2-6-2-6-2-1-5-1-5-1-5-1-5-4-3-4-3-4-3-4-3
26	Local_Best	7920	2560	29268	1-4-2-6-1-3-4-5-3-5-2-3-4-2-1-6-4-6-6-5-5-2-3-1	6-6-4-6-4-6-4-2-4-2-2-2-3-1-3-1-1-1-3-5-3-5-5-5

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
7	Local_Best	1866	80	17835	5-4-4-2-3-2-1-3-2-2-4-5-1-5-3-4-3-5-1-1	1-1-1-1-2-2-2-2-4-4-4-4-3-3-3-3-5-5-5-5
9	Local_Best	2254	80	1997	1-3-1-2-5-3-2-4-1-2-4-4-2-1-3-5-3-5-4-5	3-3-3-3-2-2-2-2-5-5-5-5-1-1-1-1-4-4-4-4
8	Local_Best	2272	80	18176	4-5-4-5-1-4-3-3-2-1-2-5-3-2-1-3-4-1-2-5	3-3-3-3-4-4-4-4-5-5-5-5-1-1-1-1-2-2-2-2
7	Local_Best	1838	80	15036	4-4-2-1-5-1-3-3-4-2-4-5-2-5-3-1-2-1-3-5	1-1-1-1-2-2-2-2-3-3-3-3-4-4-4-4-5-5-5-5
7	Local_Best	2240	80	15693	3-4-5-3-1-2-2-5-3-5-5-1-2-4-2-3-4-1-1-4	2-2-2-2-1-1-1-1-4-4-4-4-3-3-3-3-5-5-5-5
9	Local_Best	2254	80	20337	2-1-2-3-4-3-5-5-5-1-4-1-1-3-4-5-4-2-2-3	3-3-3-3-4-4-4-4-1-1-1-1-2-2-2-2-5-5-5-5

9	Local_Best	2240	80	20473	4-3-2-3-1-3-2-5-1-4-1-1-2-5-2-5-4-3-5-4	1-1-1-1-3-3-3-3-5-5-5-5-4-4-4-4-2-2-2-2
7	Local_Best	1268	80	16202	4-3-3-4-2-5-5-1-5-1-2-4-3-5-1-3-2-4-2-1	1-1-1-1-2-2-2-2-5-5-5-5-4-4-4-4-3-3-3-3
8	Local_Best	2648	80	17437	2-2-3-1-2-3-5-4-5-4-5-5-3-2-4-3-1-1-1-4	1-1-1-1-3-3-3-3-4-4-4-4-2-2-2-2-5-5-5-5
6	Local_Best	1636	80	12942	1-2-4-3-3-3-4-2-5-1-5-5-5-4-2-1-1-4-2-3	5-5-5-5-1-1-1-1-2-2-2-2-4-4-4-4-3-3-3-3

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
11	Local_Best	2769	737	31192	4-2-1-3-1-5-5-2-5-4-3-3-1-1-3-2-4-4-2-5	3-3-1-1-3-3-1-1-5-5-5-5-2-2-4-4-2-2-4-4
9	Local_Best	2917	705	19156	1-5-5-3-2-2-4-1-5-1-4-3-5-4-3-1-2-4-2-3	4-4-1-1-4-4-2-2-1-1-2-2-3-3-5-5-3-3-5-5
5	Local_Best	1921	737	10795	1-4-1-2-1-2-4-1-2-3-2-4-3-3-5-3-5-4-5-5	2-2-5-5-2-2-5-5-3-3-3-3-1-1-4-4-1-1-4-4
7	Local_Best	1951	765	14237	4-2-4-3-5-1-5-4-1-3-1-5-3-1-5-2-2-2-3-4	1-1-2-2-1-1-2-2-4-4-3-3-4-4-5-5-3-3-5-5
8	Local_Best	1889	677	16614	4-3-4-2-5-2-3-2-4-2-3-1-3-5-1-1-1-5-4-5	1-1-1-1-4-4-3-3-4-4-3-3-5-5-2-2-5-5-2-2
8	Local_Best	2895	677	16562	1-4-3-2-3-1-4-1-1-3-2-2-5-5-2-5-5-3-4-4	3-3-3-3-1-1-4-4-1-1-4-4-5-5-2-2-5-5-2-2
8	Local_Best	2163	765	16716	3-2-1-1-1-1-4-4-3-4-3-5-2-2-5-5-4-2-3-5	5-5-2-2-5-5-2-2-1-1-3-3-1-1-4-4-3-3-4-4

6	Local_Best	2703	765	129	2-5-3-1-4-3-4-2-4-4-1-5-1-5-1-5-3-2-3-2	5-5-2-2-5-5-2-2-4-4-1-1-4-4-3-3-1-1-3-3
8	Local_Best	2637	705	17018	5-4-5-1-2-3-1-3-4-2-1-2-2-4-3-4-1-3-5-5	4-4-3-3-4-4-5-5-3-3-5-5-1-1-2-2-1-1-2-2
7	Local_Best	2357	705	14943	3-5-4-5-4-1-2-3-5-4-4-1-2-2-3-2-1-1-3-5	2-2-1-1-2-2-3-3-1-1-3-3-5-5-4-4-5-5-4-4

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
11	Local_Best	7240	2560	34558	6-4-6-1-2-4-4-4-1-5-3-1-6-3-1-3-6-2-5-2-2-5-3-5	6-1-6-1-6-1-6-1-5-5-5-5-4-4-3-4-3-4-3-2-3-2-2-2
11	Local_Best	7720	2560	36259	4-1-5-2-1-2-3-5-4-1-6-2-6-2-6-3-5-1-3-5-4-6-3-4	1-1-1-1-4-3-4-3-4-3-4-3-6-2-6-2-6-2-6-2-5-5-5-5
10	Local_Best	7240	720	73012	5-5-4-3-6-3-4-4-5-1-5-1-2-6-3-2-1-3-4-2-1-6-6-2	2-5-2-5-2-5-2-5-1-6-1-6-1-6-1-6-3-4-3-4-3-4-3-4
10	Local_Best	5480	2560	73722	5-4-2-3-4-3-2-1-6-4-4-6-3-5-6-1-6-3-5-2-2-1-1-5	5-5-5-5-3-4-3-4-4-4-3-2-3-2-6-2-6-2-6-1-6-1-1-1
9	Local_Best	5480	2560	6003	2-5-1-3-3-4-4-5-3-5-6-1-6-6-1-2-2-5-1-6-2-4-4-3	5-5-4-5-4-5-4-6-4-6-6-6-1-1-1-1-2-3-2-3-2-3-2-3
8	Local_Best	5600	2560	5735	5-2-1-4-4-5-2-4-1-3-6-2-1-5-5-3-6-1-2-4-3-6-3-6	2-3-2-3-2-3-2-3-6-6-6-6-4-1-4-1-1-1-4-5-4-5-5-5
11	Local_Best	5880	2560	90371	2-3-2-3-1-2-4-6-2-4-5-5-5-6-5-6-1-1-3-4-4-1-6-3	2-2-2-2-6-6-1-6-1-6-1-4-1-4-4-4-5-3-5-3-5-3-5-3
10	Local_Best	5560	2560	75893	6-2-6-3-1-1-5-4-5-6-2-4-3-6-1-1-4-4-2-3-5-2-3-5	4-4-1-4-1-4-1-5-1-5-5-5-3-3-3-3-6-2-6-2-6-2-6-2

8	Local_Best	5760	2560	54876	4-1-2-6-5-3-4-3-6-2-6-2-4-3-5-1-6-5-4-1-2-1-3-5	4-4-1-4-1-4-1-2-1-2-2-2-3-3-3-3-6-5-6-5-6-5-6-5
12	Local_Best	5880	2560	62441	4-6-3-3-2-6-6-2-4-3-5-1-1-2-4-5-5-3-1-1-2-5-6-4	4-4-4-4-1-2-1-2-1-2-1-2-6-6-6-6-5-3-5-3-5-3-5-3

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8	Local_Best	2242	80	28761	4-5-2-2-5-3-2-4-3-2-3-3-5-1-1-4-1-4-5-1	2-2-2-2-4-4-4-4-5-5-5-5-1-1-1-1-3-3-3-3
6	Local_Best	1624	80	25318	2-1-4-1-2-5-5-3-1-3-2-4-2-1-3-5-3-5-4-4	2-2-2-2-1-1-1-1-4-4-4-4-3-3-3-3-5-5-5-5
8	Local_Best	1870	80	26012	4-1-4-3-2-5-2-3-4-3-3-5-1-5-5-2-2-4-1-1	2-2-2-2-3-3-3-3-1-1-1-1-5-5-5-5-4-4-4-4
8	Local_Best	2252	80	27042	2-4-2-4-5-2-1-4-3-3-2-5-1-5-3-1-5-4-1-3	1-1-1-1-5-5-5-5-4-4-4-4-3-3-3-3-2-2-2-2
8	Local_Best	1888	80	2829	3-3-5-2-1-1-2-3-1-1-4-5-4-4-3-2-4-2-5-5	5-5-5-5-4-4-4-4-1-1-1-1-2-2-2-2-3-3-3-3
7	Local_Best	1466	80	26097	1-4-3-3-1-1-3-5-2-2-4-5-5-2-4-3-4-1-5-2	2-2-2-2-4-4-4-4-3-3-3-3-1-1-1-1-5-5-5-5
8	Local_Best	2260	80	27686	1-5-4-5-5-5-3-3-3-2-2-3-2-4-4-1-1-2-4-1	1-1-1-1-5-5-5-5-4-4-4-4-2-2-2-2-3-3-3-3
7	Local_Best	2648	80	25282	2-4-4-5-3-1-4-2-1-5-2-3-1-3-5-3-5-2-4-1	5-5-5-5-4-4-4-4-1-1-1-1-2-2-2-2-3-3-3-3
7	Local_Best	1850	80	24168	4-5-4-4-1-3-3-2-1-3-5-4-5-1-1-3-2-2-5-2	2-2-2-2-4-4-4-4-1-1-1-1-3-3-3-3-5-5-5-5
8	Local_Best	1840	80	28111	5-3-2-3-1-5-4-4-5-5-1-2-2-3-4-4-1-3-2-1	3-3-3-3-5-5-5-5-2-2-2-2-4-4-4-4-1-1-1-1

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
8	Local_Best	2841	705	31572	4-5-2-3-4-3-3-1-5-1-5-1-2-3-4-2-5-4-1-2	3-3-4-4-3-3-2-2-4-4-2-2-1-1-5-5-1-1-5-5
9	Local_Best	2237	765	32875	5-2-1-5-3-4-3-4-1-3-3-1-5-1-2-2-2-5-4-4	4-4-3-3-4-4-3-3-2-2-1-1-2-2-5-5-1-1-5-5
10	Local_Best	2037	677	3431	3-5-4-4-4-1-2-1-3-1-5-2-2-3-1-3-5-5-2-4	4-4-4-4-1-1-3-3-1-1-3-3-2-2-5-5-2-2-5-5
8	Local_Best	2131	677	28469	3-4-3-1-4-5-4-3-5-2-2-2-4-1-2-3-1-5-5-1	5-5-5-5-3-3-2-2-3-3-2-2-4-4-1-1-4-4-1-1
8	Local_Best	2065	705	28794	4-5-3-5-4-1-1-1-3-4-2-2-5-1-5-2-4-3-2-3	4-4-5-5-4-4-3-3-5-5-3-3-2-2-1-1-2-2-1-1
8	Local_Best	2011	705	29769	1-4-3-2-3-2-1-1-1-2-4-4-2-5-3-5-3-5-4-5	3-3-4-4-3-3-5-5-4-4-5-5-2-2-1-1-2-2-1-1
8	Local_Best	1917	705	2704	3-3-5-4-5-1-1-2-3-4-1-5-3-2-2-5-4-2-4-1	2-2-4-4-2-2-5-5-4-4-5-5-3-3-1-1-3-3-1-1
8	Local_Best	2789	745	30035	1-5-2-5-3-2-5-3-4-1-5-1-2-1-2-3-3-4-4-4	3-3-5-5-3-3-2-2-4-4-2-2-4-4-1-1-5-5-1-1
6	Local_Best	1923	745	22911	1-4-1-4-2-2-1-3-3-2-5-4-3-4-5-2-3-1-5-5	4-4-3-3-4-4-5-5-2-2-5-5-2-2-1-1-3-3-1-1
8	Local_Best	2075	677	29078	4-4-1-5-2-1-1-4-2-1-2-5-3-5-3-3-2-4-5-3	1-1-1-1-2-2-4-4-2-2-4-4-5-5-3-3-5-5-3-3

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Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
10	Local_Best	5400	720	146387	3-1-4-1-3-2-5-3-5-4-2-2-4-2-6-3-5-6-1-6-5-1-4-6	1-3-1-3-1-3-1-3-2-5-2-5-2-5-2-5-4-6-4-6-4-6-4-6
9	Local_Best	5640	720	217992	3-1-4-2-2-6-6-4-6-5-5-3-1-2-3-4-5-4-2-5-1-6-1-3	4-3-4-3-4-3-4-3-2-6-2-6-2-6-2-6-5-1-5-1-5-1-5-1
11	Local_Best	5920	2560	184865	5-1-5-4-6-4-2-5-3-2-2-3-4-2-1-4-3-6-5-1-3-1-6-6	6-6-4-6-4-6-4-2-4-2-2-2-5-5-1-5-1-5-1-3-1-3-3-3
9	Local_Best	5560	2560	55877	2-2-5-1-6-1-4-1-4-5-2-3-1-6-2-3-6-5-5-4-3-4-6-3	1-5-1-5-1-5-1-5-4-4-3-4-3-4-3-6-3-6-6-6-2-2-2-2
10	Local_Best	5920	2560	63305	5-1-6-1-6-6-1-2-2-5-1-2-3-6-3-4-4-3-3-5-4-4-2-5	3-2-3-2-2-2-3-5-3-5-5-5-6-4-6-4-6-4-6-4-1-1-1-1
8	Local_Best	5320	2560	46016	3-4-5-6-2-5-5-2-1-6-2-6-2-3-6-5-1-4-4-4-1-1-3-3	4-4-6-4-6-4-6-5-6-5-5-5-2-3-3-3-2-3-2-1-2-1-1-1
10	Local_Best	5720	2560	59854	2-2-5-6-1-6-5-1-5-3-4-6-1-3-3-4-4-3-1-2-5-4-2-6	1-4-4-4-1-4-1-3-1-3-6-3-6-3-6-5-6-5-5-5-2-2-2-2
10	Local_Best	5600	2560	54108	1-4-5-6-5-4-5-3-6-5-1-1-2-3-2-3-4-1-6-4-3-6-2-2	1-1-1-1-3-2-3-2-3-2-3-2-4-4-4-4-6-5-6-5-6-5-6-5
8	Local_Best	5760	2560	45273	3-2-1-5-3-2-3-1-5-5-6-6-4-4-4-5-2-6-2-6-3-1-1-4	1-6-1-6-1-6-1-6-3-2-3-2-2-2-3-4-3-4-4-4-5-5-5-5
11	Local_Best	5600	2560	59657	3-3-2-4-2-4-4-6-1-1-3-6-5-5-6-4-1-1-5-2-6-2-3-5	5-4-5-4-5-4-5-4-6-6-6-6-1-2-1-2-1-2-1-2-3-3-3-3

Beam search (10)

basic.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State

8	Local_Best	2064	80	54975	4-2-4-5-5-5-4-5-1-3-2-2-1-3-2-1-3-1-4-3	5-5-5-5-1-1-1-1-4-4-4-4-3-3-3-3-2-2-2-2
8	Local_Best	2080	80	65676	4-1-4-4-3-1-1-5-5-2-1-4-3-5-2-2-2-3-5-3	1-1-1-1-5-5-5-5-3-3-3-3-4-4-4-4-2-2-2-2
7	Local_Best	1864	80	51268	3-3-5-4-1-2-1-1-1-2-3-4-4-2-5-4-5-3-2-5	3-3-3-3-4-4-4-4-2-2-2-2-5-5-5-5-1-1-1-1
8	Local_Best	1874	80	70171	5-2-2-2-2-1-3-5-5-4-4-5-3-1-4-1-4-1-3-3	1-1-1-1-2-2-2-2-5-5-5-5-3-3-3-3-4-4-4-4
7	Local_Best	1614	80	51029	5-3-1-5-3-4-1-1-2-1-5-4-5-2-2-4-3-2-4-3	5-5-5-5-1-1-1-1-2-2-2-2-3-3-3-3-4-4-4-4
8	Local_Best	1848	80	62585	1-1-5-3-3-5-2-3-4-5-4-4-3-4-2-5-1-2-1-2	5-5-5-5-3-3-3-3-1-1-1-1-2-2-2-2-4-4-4-4
7	Local_Best	1628	80	5023	1-5-5-3-3-2-5-1-2-2-2-4-4-3-4-5-1-4-3-1	5-5-5-5-3-3-3-3-4-4-4-4-1-1-1-1-2-2-2-2
7	Local_Best	1864	80	47328	3-3-2-5-4-2-4-4-5-4-3-2-2-5-1-1-1-1-5-3	5-5-5-5-1-1-1-1-3-3-3-3-2-2-2-2-4-4-4-4
7	Local_Best	2034	80	4965	2-2-1-1-4-5-3-5-2-4-3-5-1-4-5-4-3-3-2-1	1-1-1-1-5-5-5-5-4-4-4-4-2-2-2-2-3-3-3-3
7	Local_Best	2058	80	52976	5-4-4-4-1-2-3-5-2-1-4-3-1-1-3-2-2-3-5-5	4-4-4-4-1-1-1-1-2-2-2-2-3-3-3-3-5-5-5-5

trade.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
7	Local_Best	1985	705	5722	3-2-4-5-5-1-3-1-5-3-4-4-5-1-2-2-4-3-2-1	3-3-2-2-3-3-5-5-2-2-5-5-4-4-1-1-4-4-1-1

7	Local_Best	1897	705	64815	3-5-1-3-4-1-5-1-4-5-4-4-2-2-1-2-3-2-3-5	2-2-5-5-2-2-1-1-5-5-1-1-4-4-3-3-4-4-3-3
6	Local_Best	2111	737	50063	1-4-5-4-3-5-2-2-4-3-5-4-2-3-3-1-1-2-1-5	3-3-4-4-3-3-4-4-5-5-5-5-1-1-2-2-1-1-2-2
7	Local_Best	2123	705	51756	5-4-4-2-1-2-4-5-5-3-5-2-1-2-1-1-4-3-3-3	4-4-5-5-4-4-3-3-5-5-3-3-1-1-2-2-1-1-2-2
7	Local_Best	1997	705	50483	5-4-1-1-2-2-3-2-3-5-4-2-4-1-4-5-5-3-3-1	2-2-3-3-2-2-5-5-3-3-5-5-1-1-4-4-1-1-4-4
6	Local_Best	1925	677	44154	1-5-3-5-3-3-2-2-5-2-2-4-1-4-3-5-4-1-4-1	4-4-4-4-2-2-5-5-2-2-5-5-3-3-1-1-3-3-1-1
8	Local_Best	2125	705	61953	4-2-5-2-3-2-5-4-4-1-5-1-4-3-3-5-2-1-1-3	5-5-2-2-5-5-1-1-2-2-1-1-3-3-4-4-3-3-4-4
6	Local_Best	1743	677	41964	4-2-4-4-2-5-1-3-5-1-3-4-1-3-5-5-2-3-2-1	1-1-1-1-4-4-5-5-4-4-5-5-2-2-3-3-2-2-3-3
7	Local_Best	2215	677	495	2-5-1-1-5-4-2-4-3-4-2-2-5-1-4-3-3-3-1-5	4-4-4-4-2-2-3-3-2-2-3-3-1-1-5-5-1-1-5-5
6	Local_Best	1791	677	42812	4-3-2-4-1-5-4-3-2-5-2-1-1-5-5-2-3-3-1-4	5-5-5-5-2-2-4-4-2-2-4-4-3-3-1-1-3-3-1-1

trap.txt

Iterations	End_Reason	Initial_Eval	Final_Eval	Runtime	Initial_State	Final_State
8	Local_Best	5440	2560	302658	2-1-6-3-5-1-3-2-4-1-4-6-5-2-1-5-2-3-3-4-6-5-6-4	2-2-2-2-1-6-1-6-6-6-1-3-1-3-3-3-5-4-5-4-5-4-5-4
10	Local_Best	5400	720	418455	3-1-6-4-2-4-5-5-1-3-3-4-2-2-6-1-4-5-6-6-3-5-1-2	2-3-2-3-2-3-2-3-1-4-1-4-1-4-1-4-6-5-6-5-6-5-6-5

11	Local_Best	5400	720	116992	4-6-2-1-1-5-5-2-3-5-6-5-4-3-4-3-2-6-1-6-4-3-1-2	5-2-5-2-5-2-5-2-3-1-3-1-3-1-3-1-6-4-6-4-6-4-6-4
6	Local_Best	5720	720	64915	5-1-3-6-1-3-6-2-1-5-1-4-2-5-4-5-3-2-4-3-2-4-6-6	4-6-4-6-4-6-4-6-3-5-3-5-3-5-3-5-2-1-2-1-2-1-2-1
10	Local_Best	5640	2560	106745	4-1-3-2-3-5-4-6-1-1-4-5-6-5-3-1-6-3-2-2-2-4-6-5	5-2-5-2-2-2-5-3-5-3-3-3-1-4-1-4-1-4-1-4-6-6-6-6
10	Local_Best	5200	720	109729	6-6-6-5-5-1-2-5-1-3-4-1-2-3-3-1-5-2-4-2-6-4-4-3	2-6-2-6-2-6-2-6-3-5-3-5-3-5-3-5-1-4-1-4-1-4-1-4
10	Local_Best	5600	720	108503	2-3-1-4-3-4-1-6-6-4-4-5-5-6-2-1-2-6-1-2-5-5-3-3	2-6-2-6-2-6-2-6-4-3-4-3-4-3-4-3-1-5-1-5-1-5-1-5
9	Local_Best	7320	2560	95767	4-1-1-6-6-5-5-3-4-2-6-3-6-4-2-4-5-2-2-1-3-5-1-3	2-3-2-3-2-3-2-3-4-4-1-4-1-4-1-6-1-6-6-6-5-5-5-5
11	Local_Best	5560	720	118706	6-2-4-3-2-5-3-1-1-4-5-3-4-6-4-1-1-2-5-3-2-5-6-6	6-1-6-1-6-1-6-1-3-4-3-4-3-4-3-4-5-2-5-2-5-2-5-2
9	Local_Best	5680	2560	96047	4-5-2-2-6-5-4-5-6-1-3-5-3-4-1-6-2-2-6-1-3-3-1-4	6-4-6-4-6-4-6-4-5-5-5-5-2-2-2-2-1-3-1-3-1-3-1-3

2.2. Comparación de Resultados

Compara los resultados obtenidos por las estrategias de búsqueda y razona el comportamiento de los algoritmos de acuerdo a sus características teóricas. Para guiar tu análisis, asegúrate de abordar los siguientes puntos en tu redacción:

- **Eficacia vs. Eficiencia**: Observando los resultados sobre basic.txt, apreciamos que no es siempre necesario recurrir a algoritmos complejos. Tanto el Hill Climbing básico como sus variantes más pesadas (Random Restart, Beam Search) logran hallar el mismo coste óptimo garantizado. Dado que el espacio de búsqueda es sencillo y directo hacia el óptimo global el Hill Climbing clásico resulta ser la mejor elección por el balance entre eficacia y un tiempo menor al no realizar re-evaluaciones innecesarias ni gestionar la memoria de múltiples ramas.
- **Óptimos locales y "landscape" engañoso (trap.txt)**: El Hill Climbing se viene bastante abajo con la instancia trap.txt: casi siempre acaba en soluciones malas o mediocres y no consigue llegar al 720. Básicamente pasa porque el problema está diseñado de forma "tramposa": la heurística te empuja

hacia zonas que parecen buenas al principio, pero en realidad son callejones sin salida donde todos los vecinos son peores, así que te quedas atascado en un óptimo local. Lo bueno de Random Restart es que evita este bloqueo reiniciando el estado de forma aleatoria cuando ve que no avanza. Al probar muchas veces desde puntos distintos del espacio de soluciones, aumenta bastante la probabilidad de que alguna ejecución empiece en una zona que sí lleve al óptimo global.

- **La trampa de la avaricia (El caso del Beam Search)**: En tableros con engaños de la función coste (trap.txt), Beam Search suele fracasar y comportarse peor que Random Restart. La razón teórica radica en la fuga de diversidad: dado que en Beam Search los K estados se seleccionan de la misma bolsa común de sucesores basándose puramente en valoraciones locales mejores, si una región subóptima parece momentáneamente atractiva, todos los K beams convergen y se aglomeran sobre ese mismo óptimo local. Random Restart, en cambio, obliga a las ejecuciones a ser independientes, forzando la diversidad y extendiendo la exploración fuera de estos "cuellos de botella".

- **Ley de rendimientos decrecientes**: Al comparar variantes como Random Restart y Beam Search con distintos multiplicadores (5, 15 y 30 reinicios/beams), observamos un claro efecto de retornos decrecientes en relación al coste computacional. Si bien pasar de 1 a 5 reinicios reduce la media de coste significativamente, el paso de 15 a 30 reinicios en Stochastic Random Restart triplica los tiempos de cómputo sin implicar mejoras proporcionales. Para este problema en concreto, el límite pragmático se sitúa en N=10 reinicios. Dicha parametrización encuentra un equilibrio ideal: elude la mayoría de óptimos locales con trampa, y mantiene un gasto temporal manejable e imperceptible comparado con configuraciones masivas.

3. Declaración de uso de herramientas de IA

Hemos utilizado Github Copilot para corregir el algoritmo de Beam Search y ChatGPT para entenderlo.

4. Bibliografía

- Apuntes y presentaciones teóricas de la asignatura (Algoritmos de Búsqueda Local, Metaheurísticas).
- Documentación del lenguaje R.